

MolPallette — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallette/coconut-flavordb_v5/naveja_recap

generated 2026-08-20 15:55

A • Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.3333333333333333, 'include_ring': True, 'max_cores':
core_ratio	0.3333333333333333
max_cores	4
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	4145
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	4
rdkit_version	2026.03.2
created_at	2026-08-20T15:53:47

B · Composition

quantity	value
molecules	393,066
from coconut	372,254 (94.7%)
from flavordb	20,812 (5.3%)
decompositions	1,145,758
per molecule	mean 2.91, median 3, max 4
R-group slots	1,608,324
per decomposition	mean 1.40
k >= 2	31.03% of decompositions

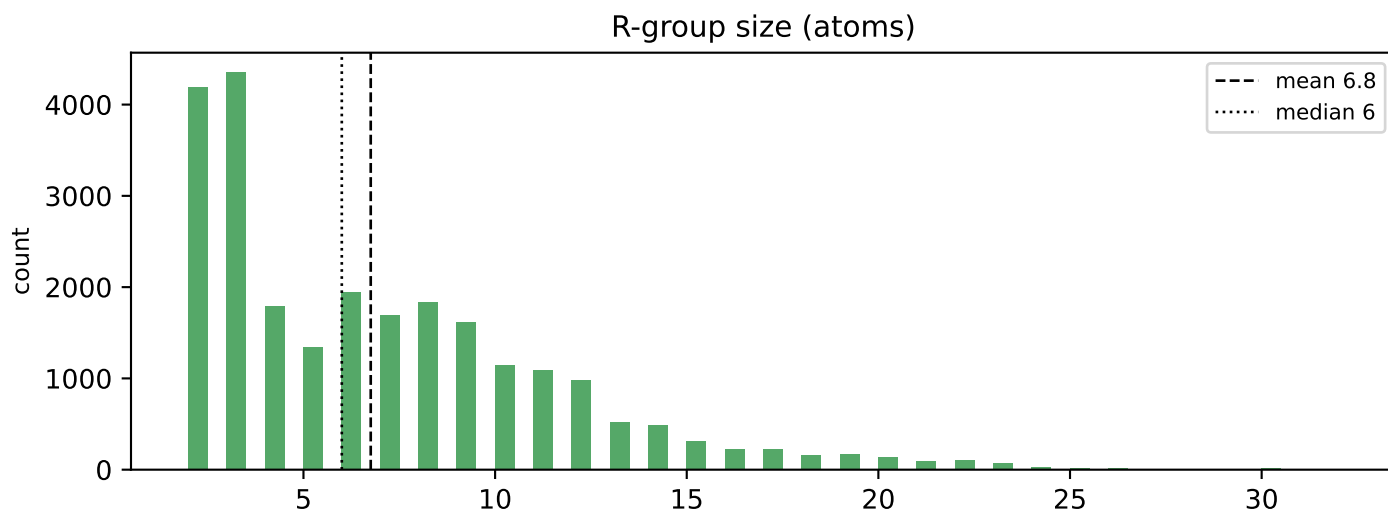
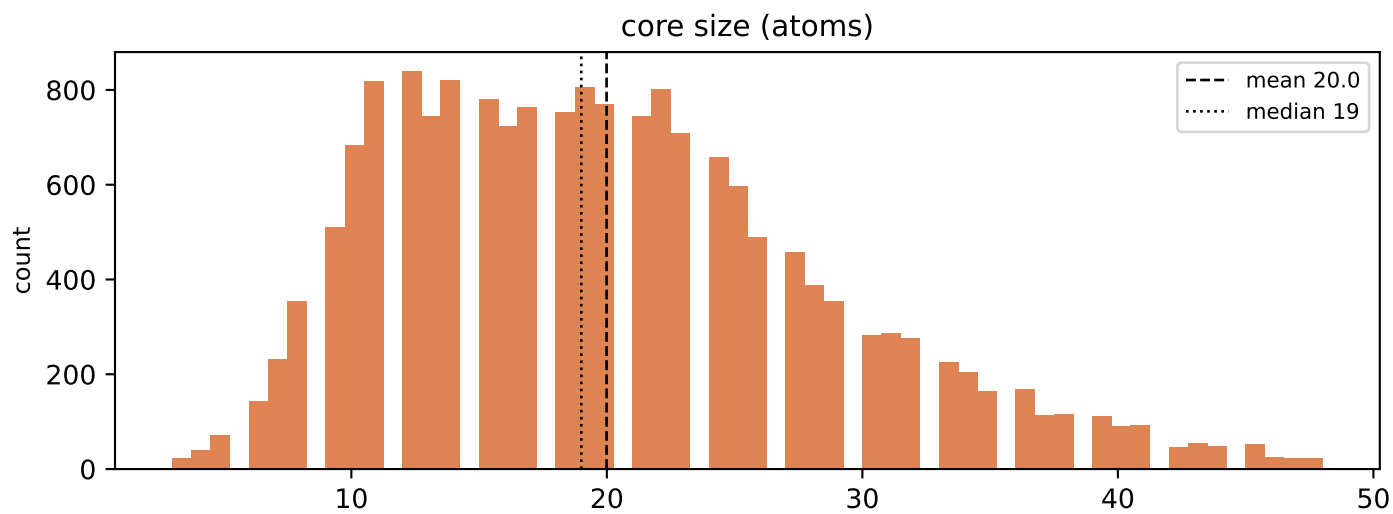
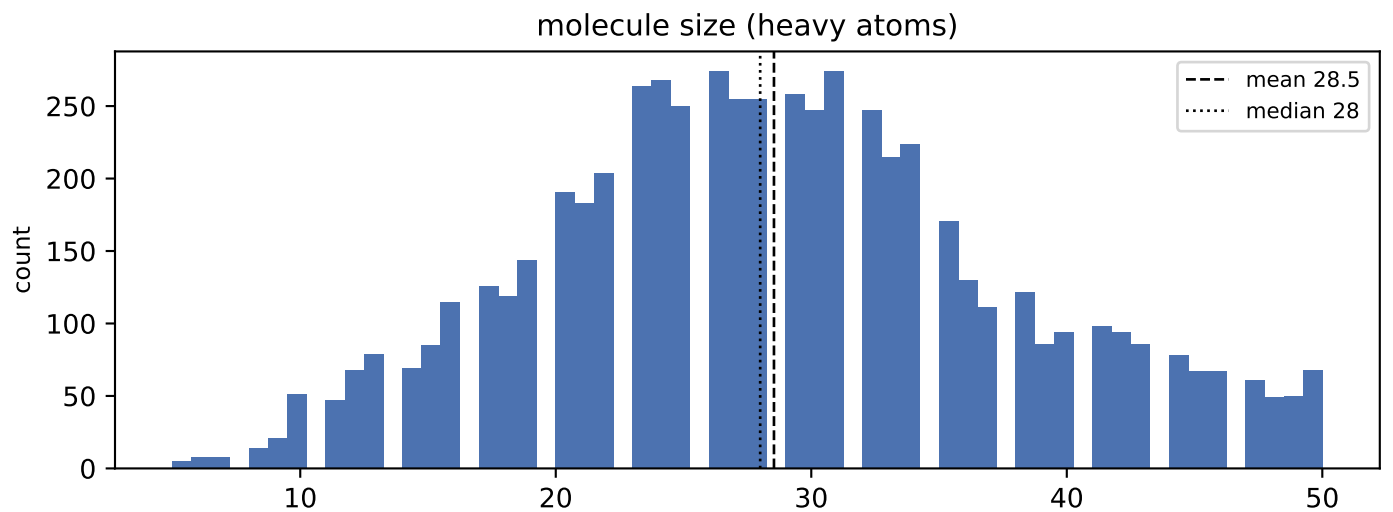
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	393,066 instances/epoch
sampling_unit = decomposition	1,145,758 instances/epoch (2.9x)
full (mol, core, islinked) space	2,508,474 (6.4 per molecule)

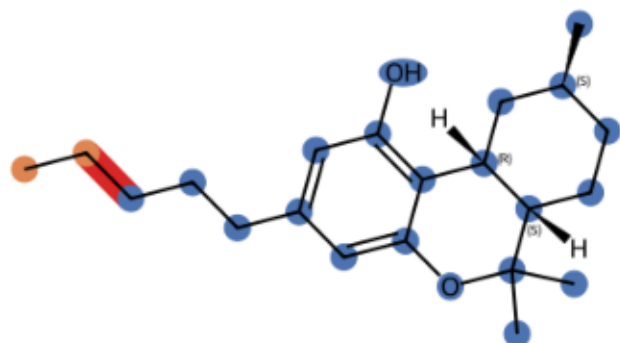
MolPLA's two stages: one molecule yields many cores; one core yields $2^k - 1$ islinked subsets.

D • Size distributions

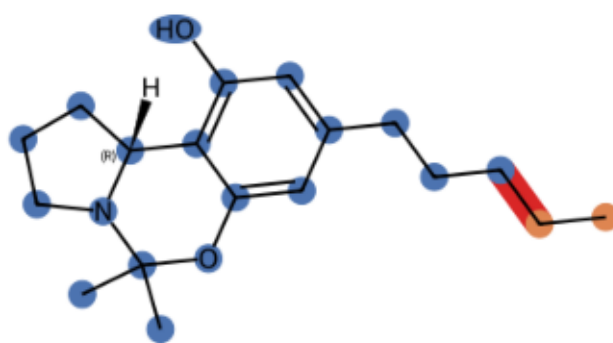


E · Example decompositions (ratio 0.3333333333333333)

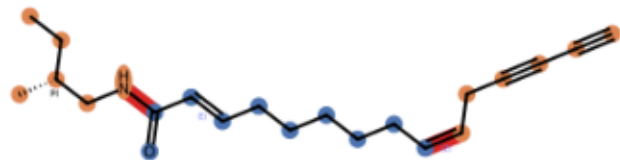
CNP0000075 k=1 core 21 atoms, R-groups [2]



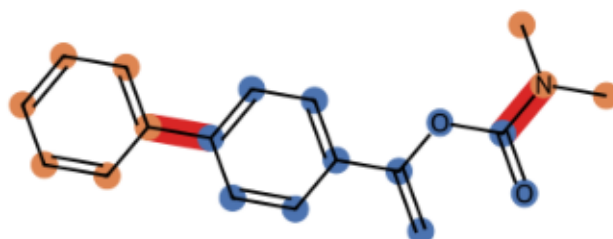
CNP0000164 k=1 core 19 atoms, R-groups [2]



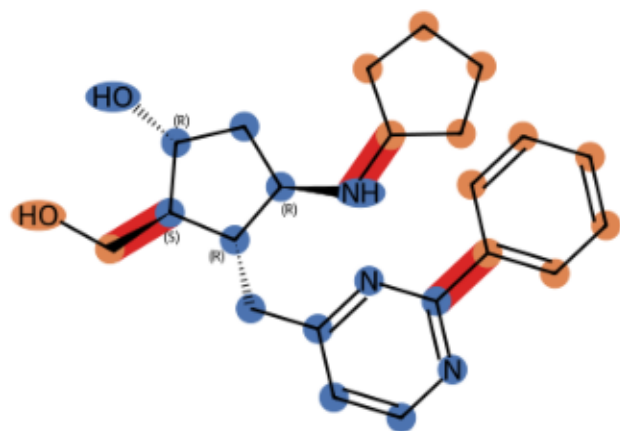
CNP0000306 k=2 core 10 atoms, R-groups [6, 6]



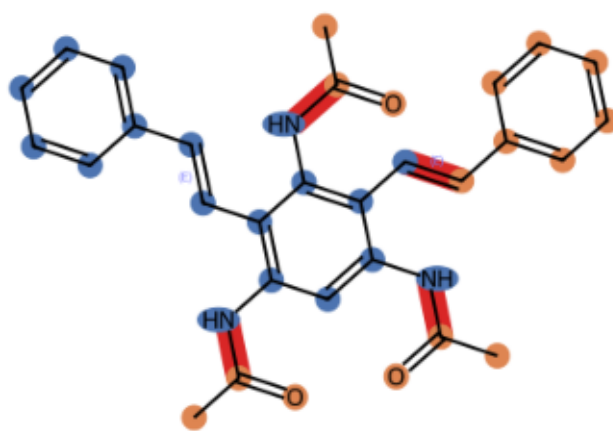
CNP0000665 k=2 core 11 atoms, R-groups [3, 6]



CNP0005977 k=3 core 14 atoms, R-groups [2, 6, 5]

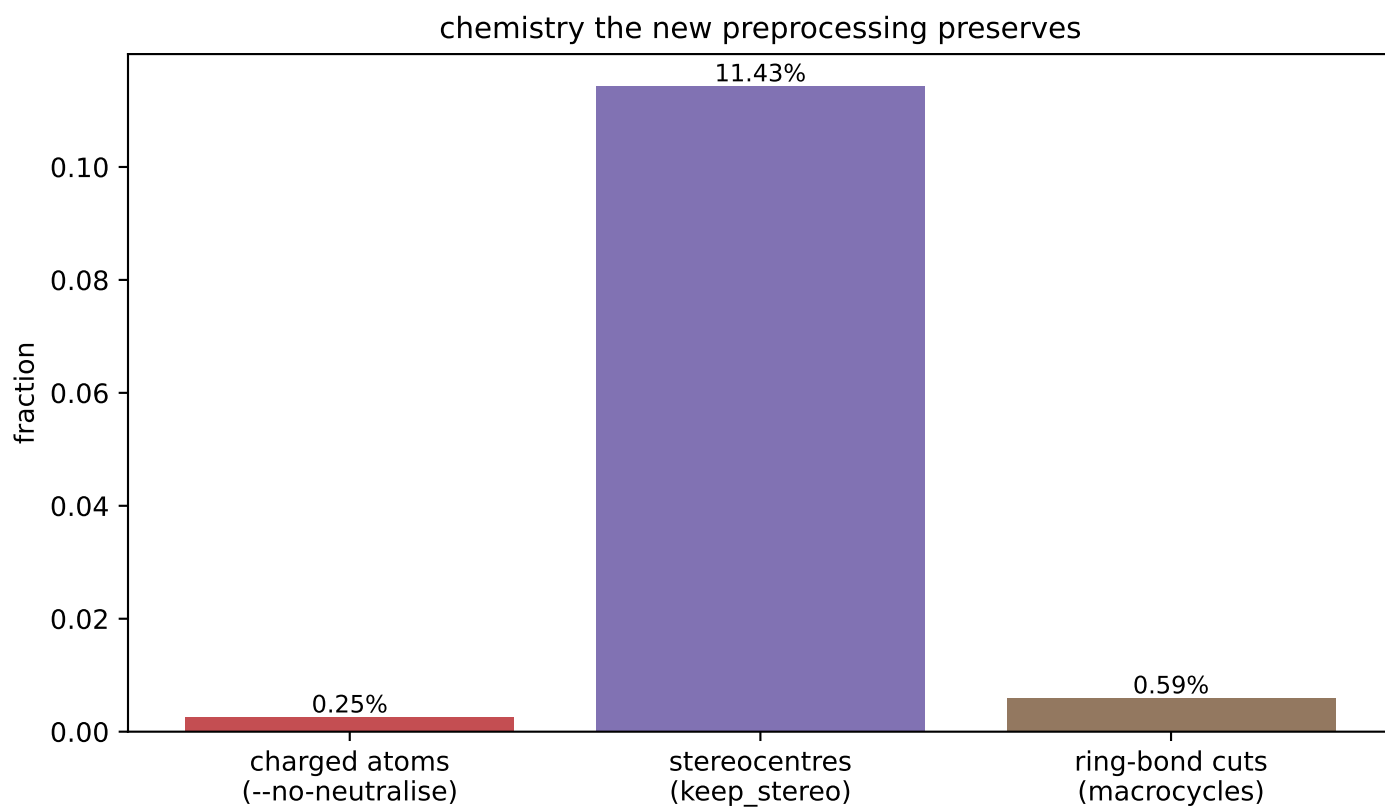
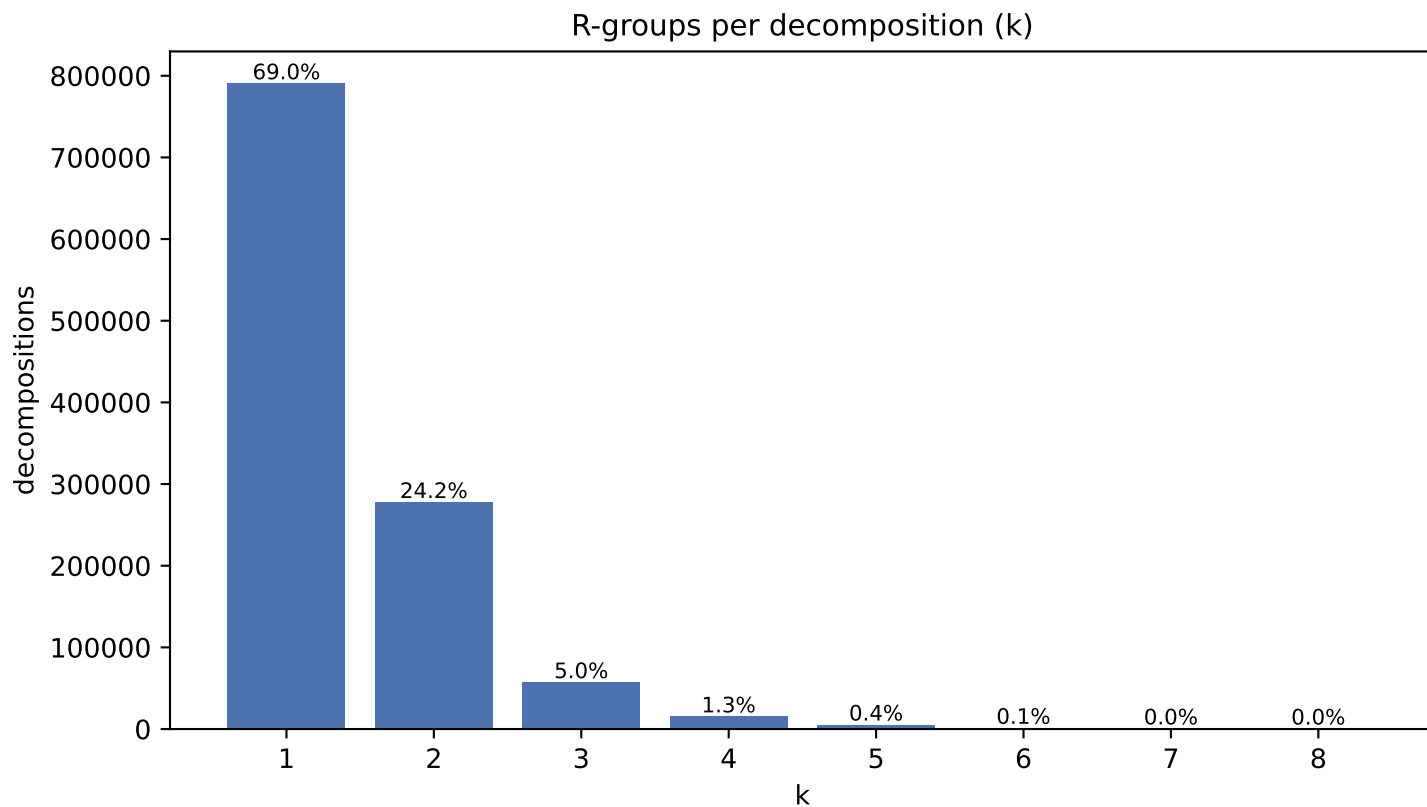


CNP0070646 k=4 core 18 atoms, R-groups [3, 3, 3, 7]

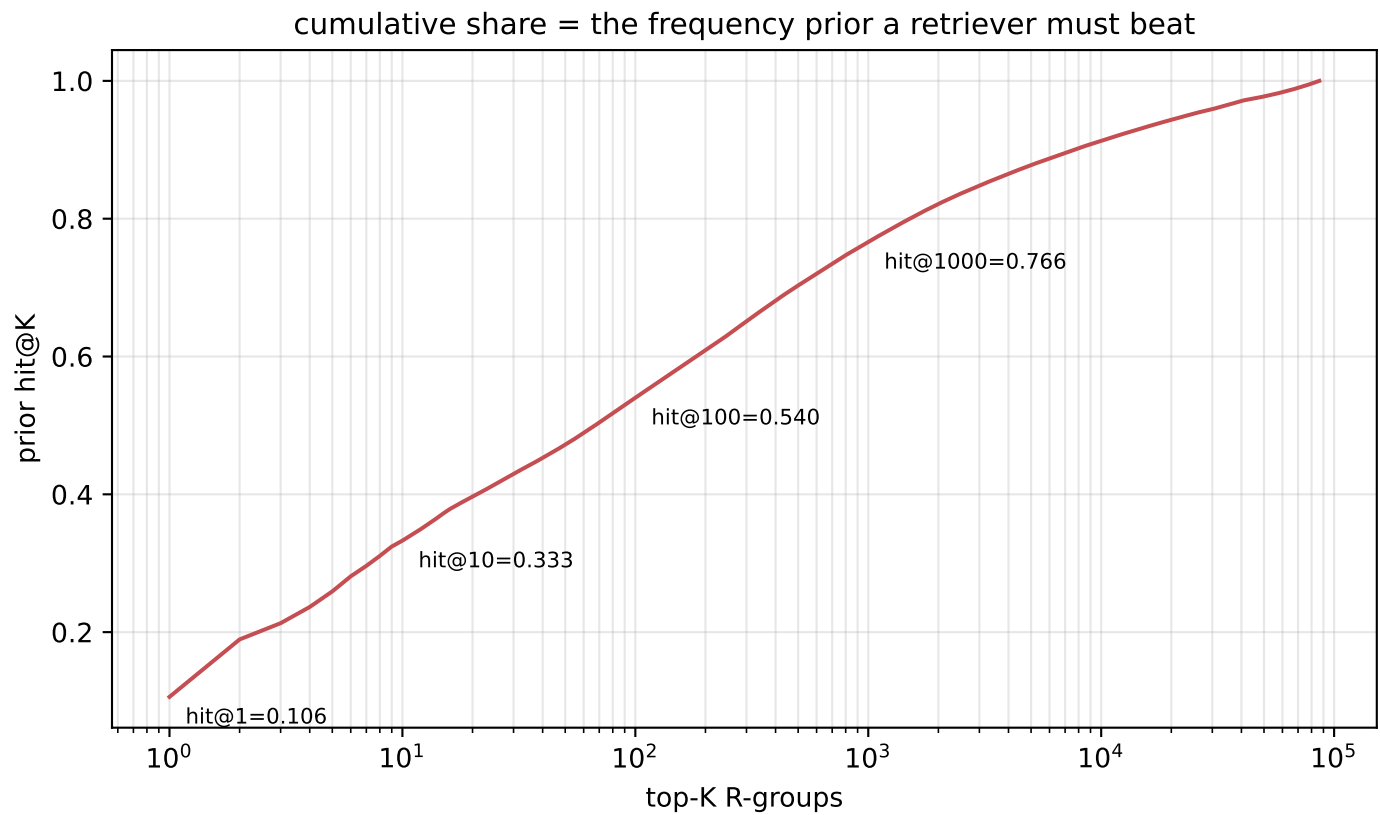
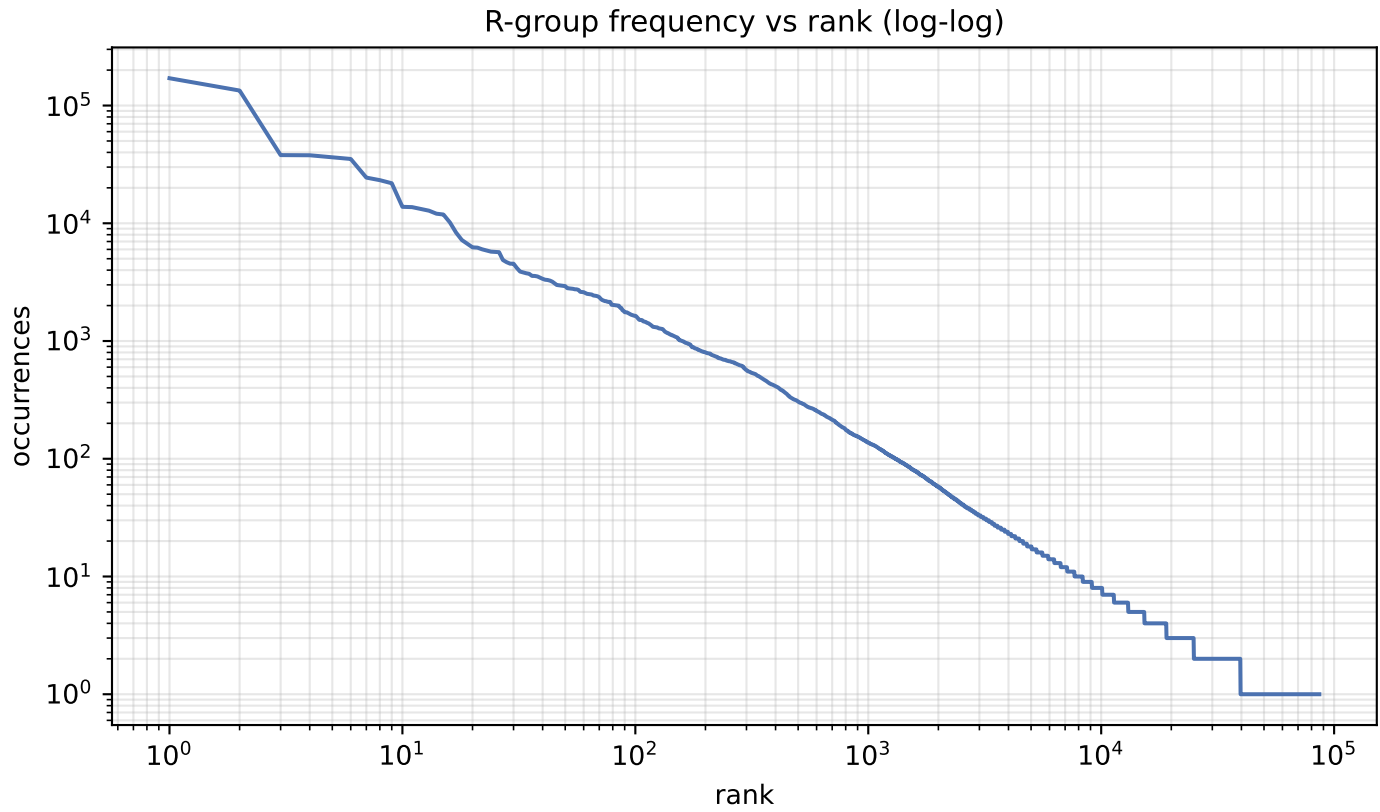


blue = core orange = R-groups red = cut bonds

F · Decomposition shape & preserved chemistry



G · R-group library skew



H · R-group library

quantity	value
distinct R-groups	86,385
occurrences	1,608,324
effective size (exp H)	889
top-1 share	10.60%
top-10 cover	33.3%
top-100 cover	54.0%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
170,561	10.60%	*OC
134,073	8.34%	*C(C)=O
37,929	2.36%	*[CH2]C
37,816	2.35%	*c1ccccc1
36,363	2.26%	*C(C)C
35,172	2.19%	*[CH2]O
24,429	1.52%	*[CH](C)C
23,237	1.44%	*OC(C)=O
21,873	1.36%	*C(=O)O
13,795	0.86%	*O[C@@H]1O[C@H](CO)[C@@H](O)[C@H](O)[C@H]1O
13,715	0.85%	*[CH2]c1ccccc1
13,253	0.82%	*[CH2]CC
12,792	0.80%	*C(=O)c1ccccc1
12,062	0.75%	*c1ccc(OC)cc1
11,866	0.74%	*OCC
10,152	0.63%	*c1ccc(O)cc1
8,332	0.52%	*[CH]=O
7,206	0.45%	*[CH2]C(=O)O
6,705	0.42%	*C1O[C@H](CO)[C@@H](O)[C@H](O)[C@H]1O
6,246	0.39%	*[CH]=C

J · Instance-level common-R-group filter

quantity	value
percentile	99.9
count threshold	1,879
hashes flagged common	87 of 86,385
their share of occurrences	52.6%
single-R-group instances rejected	12,791/24,583 (52.0%)
mean R-group size, rejected	4.03 atoms
mean R-group size, kept	9.71 atoms

Filters INSTANCES, not the library: the retrieval target space and its frequency prior are unchanged. At k=1 the rule reduces to 'drop if the R-group is common', so rejection is far heavier than MolPLA saw.